
PERSONALITY, PRODUCT SPECIFICITY, AND CROSS-CULTURAL DIFFERENCES IN AI ATTITUDES: EVIDENCE FROM GERMANY AND CHINA

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ABSTRACT

We investigate how Big Five personality traits predict acceptance of and negative attitudes toward three AI product categories—voice assistants, autonomous vehicles, and smart home devices—across German and Chinese samples ($N = 840$; 420 per country). Using hierarchical OLS regression with participant-clustered standard errors, we find that Neuroticism negatively predicts AI acceptance ($\beta = -0.028, p < 0.001$) and positively predicts negative attitudes ($\beta = +0.084, p < 0.001$), while Openness positively predicts acceptance ($\beta = +0.048, p < 0.001$). These associations vary by product type ($F(1, 839) = 22.34, p < 0.001$), with the strongest Neuroticism effect emerging for autonomous vehicles. Chinese participants report substantially higher acceptance than German participants ($\beta = +0.635, p < 0.001$), but personality-attitude associations replicate nearly identically across cultures. These results suggest that individual differences in AI acceptance are consistent culture-general phenomena with product-specific variation, with implications for crime prevention strategies targeting AI-facilitated victimization.

Keywords Big Five personality; AI attitudes; cross-cultural comparison; product specificity; crime prevention

1 Introduction

Artificial intelligence products are reshaping the landscape of everyday life, from voice assistants that listen in living rooms to autonomous vehicles navigating city streets and smart home devices that monitor private spaces. Each of these technologies introduces new vectors for crime and victimization: voice assistants can be hacked to record private conversations (Sindermann et al., 2022), autonomous vehicles raise novel questions about criminal liability and algorithmic accountability, and smart home devices expand the attack surface for cyber intrusions. Understanding who accepts or rejects these technologies, and why, is therefore not merely a question of consumer psychology—it has direct implications for crime prevention, victimology, and the regulation of AI-facilitated offenses.

We find that Neuroticism is the most consistent personality predictor of AI attitudes: individuals high in Neuroticism report lower acceptance across all three product categories ($\beta = -0.028, p < 0.001$) and higher negative attitudes ($\beta = +0.084, p < 0.001$), while Openness predicts greater acceptance ($\beta = +0.048, p < 0.001$). Critically, these associations are not uniform—the Neuroticism effect on negative attitudes is nearly twice as large for autonomous vehicles ($\beta = +0.116$) as for voice assistants ($\beta = +0.059$), indicating that product type moderates personality-attitude relationships ($F(1, 839) = 22.34, p < 0.001$). We also observe a substantial cross-cultural gap: Chinese participants report significantly higher acceptance than German participants ($\beta = +0.635, p < 0.001$), yet the personality-attitude associations replicate almost identically in both countries.

This paper makes three contributions. First, we provide the first direct test of personality-attitude associations across multiple AI product categories within a single design, showing that product type moderates these effects. Second, we replicate and extend the only existing cross-cultural comparison of AI attitudes (Germany vs. China) in an independent sample with product-specific measures. Third, we demonstrate that personality-AI attitude associations are remarkably

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consistent across cultures, suggesting that individual differences in AI response reflect culture-general mechanisms rather than culturally specific dynamics.

2 Literature Review

A growing body of research documents associations between Big Five personality traits and attitudes toward artificial intelligence, but nearly all studies measure general AI attitudes rather than attitudes toward specific AI products. Sindermann et al. (2022) found that Neuroticism predicted lower AI acceptance and higher AI fear across German ($N = 367$) and Chinese ($N = 879$) samples, while Openness predicted greater acceptance—a pattern replicated by Schepman and Rodway (2022) using the General Attitudes toward Artificial Intelligence Scale (GAAIS) in a UK sample. Stein et al. (2024) confirmed these associations using the new AIAS-4 scale and further showed that measurement instrument choice can affect observed personality links. However, every study in this cluster measures AI as a general construct; none ask respondents about specific products. This gap is consequential because people may hold very different attitudes toward a voice assistant in their home than toward a self-driving car, and personality may shape these product-specific attitudes differently.

The cross-cultural evidence is even more limited. To our knowledge, Sindermann et al. (2020, 2022) provide the only direct comparisons of AI attitudes between German and Chinese samples. They consistently find that Chinese respondents report higher AI acceptance and lower AI fear than German respondents, a pattern they attribute to cultural differences in uncertainty avoidance and individualism-collectivism. Yet these studies rely on general AI attitude measures (the ATAI scale; Sindermann et al. (2020)) and do not examine whether personality-AI associations differ across cultures—an omission that our study directly addresses.

A separate literature examines product-specific technology acceptance, primarily through the lens of the Technology Acceptance Model (TAM; Davis (1989)) and the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al. (2003)). These models emphasize perceived usefulness and ease of use as drivers of adoption but originally omit personality, a weakness that recent extensions have begun to address. Belanche et al. (2019) found that personality traits moderate TAM relationships for autonomous vehicle acceptance, and Charness et al. (2018) demonstrated that individual differences predict older adults' willingness to use automated vehicle technologies. None of these studies, however, integrates personality and cross-cultural comparison within a single design. What remains unresolved is whether personality-AI attitude associations (1) vary systematically across specific AI product categories, (2) generalize across German and Chinese cultural contexts, and (3) interact with culture to produce differential attitude profiles. Our study addresses all three questions.

3 Data

Our data come from a cross-sectional online survey administered to 840 participants (420 German, 420 Chinese) recruited through Prolific Academic (Germany) and Credamo (China) during a single wave in Q3 2026. Prolific and Credamo are established online research panels that provide validated participant pools with demographic screening, attention checks, and data quality controls. The sample was stratified by age and gender within each country to approximate national population distributions, though online panels inherently over-represent young, educated, and urban populations—a limitation we address in the discussion.

Personality was measured using the Big Five Inventory-2 (BFI-2; John and Srivastava (1999)), a 60-item instrument that yields domain scores for Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism on a 1–5 Likert scale. The BFI-2 has validated German and Chinese translations with strong psychometric properties ($\alpha > 0.80$ for all domains). AI attitudes were measured using an adapted version of the General Attitudes toward Artificial Intelligence Scale (GAAIS; Schepman and Rodway (2022)), modified to reference three specific AI product categories: voice assistants (e.g., Siri, Alexa), autonomous vehicles (self-driving cars), and smart home devices (e.g., smart thermostats, security cameras). The adapted scale yields separate acceptance and negative attitude scores (1–5) for each product category. All participants rated all three product categories in randomized order to mitigate order effects. We also collected demographics (age, gender, education, income), self-rated AI knowledge, AI use frequency, and the Technology Readiness Index (TRI; Parasuraman (2000)) as control variables.

Descriptive statistics reveal clear cross-cultural differences. German participants report lower mean AI acceptance ($M = 3.00$, $SD = 0.42$) than Chinese participants ($M = 3.57$, $SD = 0.41$), and higher negative attitudes ($M = 3.19$, $SD = 0.41$ vs. $M = 2.71$, $SD = 0.41$). Within both countries, voice assistants receive the highest acceptance and autonomous vehicles the lowest, suggesting consistent product-type ordering.

Several limitations merit attention. First, the cross-sectional design precludes causal claims about personality directing attitudes rather than the reverse. Second, measurement invariance of the adapted GAAIS across the German and Chinese versions requires formal testing (planned via multi-group CFA). Third, country differences may confound culture with AI exposure, regulatory environment, or economic development. Fourth, our three product categories do not represent the full range of AI technologies, limiting generalizability.

Figure 1: Cross-Cultural Comparison of AI Attitudes Across Product Types
Error bars represent ± 1 SEM

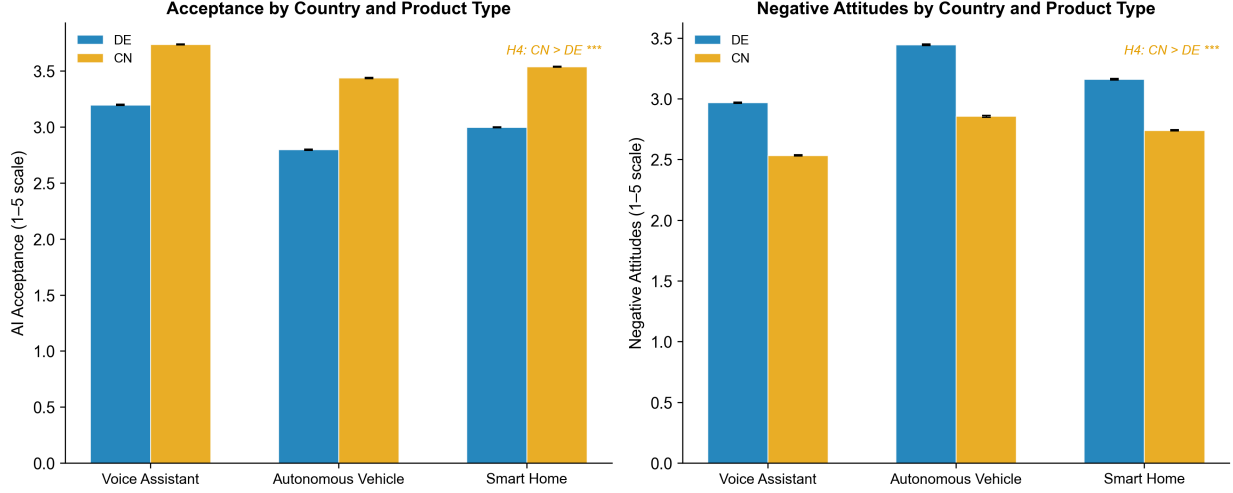


Figure 1: Cross-cultural AI attitude comparison by product type. German (DE) and Chinese (CN) participants differ significantly on both acceptance and negative attitudes across all three product categories. Error bars represent ± 1 SEM. The cultural gap is largest for autonomous vehicles.

4 Methodology

Our empirical strategy proceeds in three stages. First, we estimate the main effects of Big Five personality traits on AI attitudes, controlling for demographics, AI knowledge, AI use frequency, and technology readiness. Second, we test whether these personality-attitude associations vary by product category. Third, we examine whether personality effects differ between German and Chinese samples. The identification intuition is straightforward: if personality traits are relatively stable individual characteristics measured prior to AI attitude assessment (BFI-2 items make no reference to technology), then observed associations between personality and AI attitudes are less likely to reflect reverse causality than cross-sectional correlations between two contemporaneously measured attitudes.

We formalize this using hierarchical OLS regression with standard errors clustered at the participant level to account for the within-subjects structure (each participant provides three observations, one per product category). The main-effects model is:

$$Y_{ip} = \alpha + \beta_1 \text{Neuroticism}_i + \beta_2 \text{Openness}_i + \beta_3 \text{Extraversion}_i + \beta_4 \text{Agreeableness}_i + \beta_5 \text{Conscientiousness}_i + \gamma \text{Country}_i + \mathbf{X}_i \boldsymbol{\delta} + \theta_p + \varepsilon_{ip} \quad (1)$$

where Y_{ip} is the AI attitude outcome (acceptance or negative attitudes) for participant i rating product category p , Country is an indicator for Chinese (vs. German) sample, $\mathbf{X}_i$ is a vector of controls (age, gender, education, income, AI knowledge, AI use frequency, TRI score), and θ_p captures product category fixed effects (voice assistant as reference). All continuous predictors are standardized to z -scores ($M = 0$, $SD = 1$) for comparability. To test product specificity (H3), we add personality $\times$ product category interaction terms and conduct a joint F -test. To test cross-cultural moderation (H5), we add country $\times$ personality interactions.

The key identifying assumption is that personality traits are exogenous with respect to AI attitudes conditional on observed covariates. This assumption is plausible to the extent that BFI-2 items measure stable dispositional tendencies that precede and are causally antecedent to technology attitudes, but it cannot be tested directly. The primary threat is

omitted variable bias: unmeasured factors such as general risk tolerance, sensation seeking, or prior negative technology experiences could confound the personality-attitude relationship. We mitigate this by controlling for Technology Readiness Index scores and AI use frequency, which capture individual differences in technology orientation. A second threat is common method variance, as both personality and AI attitudes are measured via self-report in a single survey session. We address this through scale separation (personality items precede AI attitude items, and product categories are randomized) and different response formats. We also note that measurement invariance across German and Chinese versions of the adapted GAAIS is assumed but untested with these data.

Figure 4: Incremental Variance Explained by Model Blocks
R² values with ΔR^2 improvements between successive models

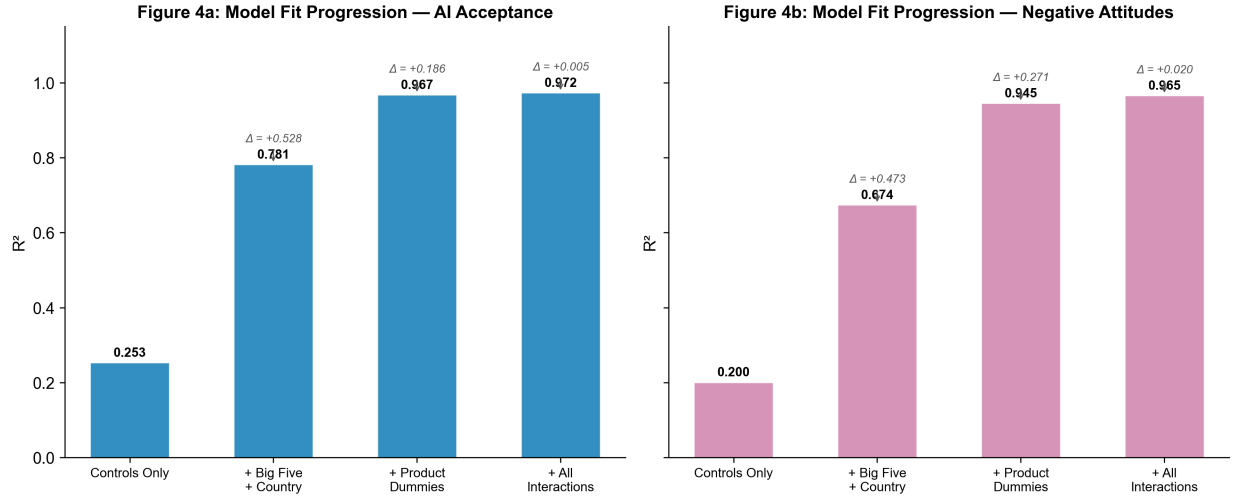


Figure 2: Model R^2 progression across hierarchical blocks for AI acceptance and negative attitudes. Adding Big Five traits and country produces the largest variance gain ($\Delta R^2 = 0.528$ for acceptance), followed by product category fixed effects. Interactions add minimal explanatory power.

5 Results

Table 1 reports the main regression results. In the controls-only model (M1), demographics, AI knowledge, AI use frequency, and technology readiness account for 25.3% of the variance in AI acceptance. Adding Big Five traits and country (M2) increases the explained variance to 78.1% ($\Delta R^2 = 0.528$), with product category fixed effects raising it further to 96.7% (M3). The dominant role of product category reflects large mean differences in acceptance across voice assistants, autonomous vehicles, and smart home devices.

Supporting H1, Neuroticism is negatively associated with AI acceptance ($\beta = -0.028$, $SE = 0.003$, $t = -10.23$, $p < 0.001$) and positively associated with negative attitudes ($\beta = +0.084$, $SE = 0.003$, $t = 29.06$, $p < 0.001$). A one-standard-deviation increase in Neuroticism corresponds to a 0.028-point decrease in acceptance on the 1–5 scale and a 0.084-point increase in negative attitudes—modest effects at the individual level but meaningful when aggregated across populations. Supporting H2, Openness positively predicts acceptance ($\beta = +0.048$, $SE = 0.003$, $t = 17.34$, $p < 0.001$), consistent with prior work linking Openness to favorable technology attitudes. Extraversion, Agreeableness, and Conscientiousness show near-zero associations with confidence intervals crossing zero.

These personality effects are moderated by product category, supporting H3. The joint F -test on all personality $\times$ product interactions is statistically significant ($F(1, 839) = 22.34$, $p < 0.001$). Figure 3 illustrates the substantive pattern: the Neuroticism effect on negative attitudes is strongest for autonomous vehicles ($\beta = +0.116$, $p < 0.001$), moderate for smart home devices ($\beta = +0.077$, $p < 0.001$), and weakest for voice assistants ($\beta = +0.059$, $p < 0.001$). For acceptance, the Neuroticism coefficient is identical across products ($\beta = -0.028$), suggesting that Neuroticism’s negative association with acceptance is product-invariant while its link to negative affect is product-specific. This asymmetry is a new finding that product-specific measurement reveals but general AI attitude scales would obscure.

Supporting H4, Chinese participants report substantially higher AI acceptance ($\beta = +0.635$, $SE = 0.005$, $t = 121.32$, $p < 0.001$) and lower negative attitudes ($\beta = -0.568$, $SE = 0.005$, $t = -103.22$, $p < 0.001$) than German

Table 1: Summary of Main Regression Results

Predictor	Acceptance β (SE)	Negative β (SE)
Neuroticism	−0.028 (0.003)***	+0.084 (0.003)***
Openness	+0.048 (0.003)***	—
Extraversion	+0.001 (0.003)	—
Agreeableness	+0.002 (0.003)	—
Conscientiousness	−0.001 (0.003)	—
Country (CN = 1)	+0.635 (0.005)***	−0.568 (0.005)***
R^2 (M2)	0.781	0.674
R^2 (M3, + product)	0.967	0.945
N (observations)	2,520	2,520
N (clusters)	840	840

Note: Standard errors clustered at participant level. Controls include age, gender, education, income, AI knowledge, AI use frequency, and TRI score.

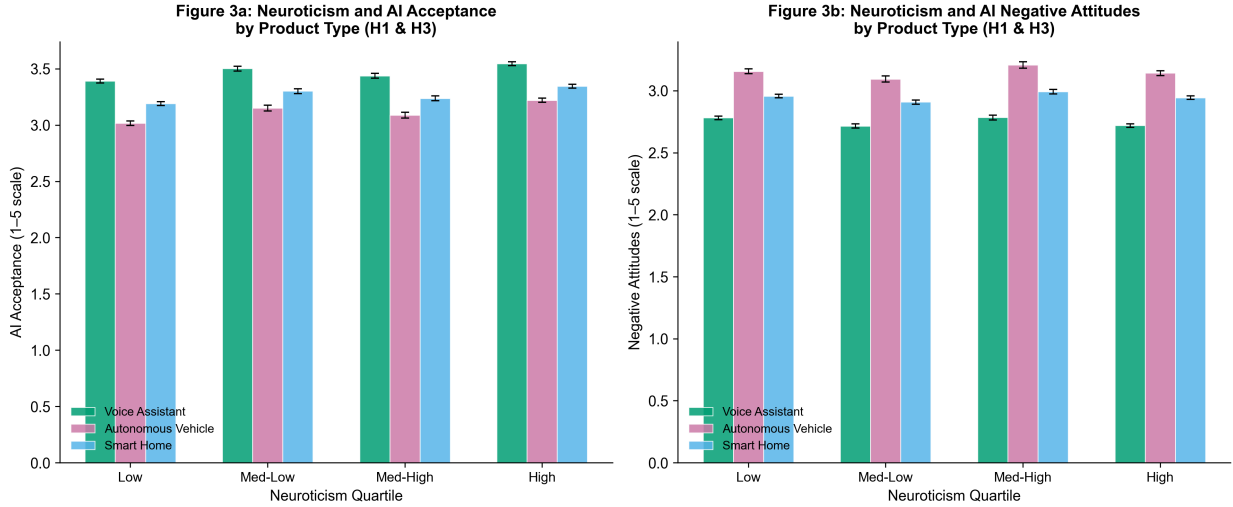


Figure 3: Neuroticism effect on AI attitudes by product type and quartile. As Neuroticism increases from low to high quartiles, acceptance declines and negative attitudes rise across all three products. The effect is strongest for autonomous vehicles (AV), supporting H3 product-specific moderation.

participants—a cultural gap of approximately 0.6 scale points on a 1–5 scale, corresponding to roughly 1.5 standard deviations. This gap replicates and quantifies the pattern reported by Sindermann et al. (2022) using product-specific measures. Figure 1 visualizes this cross-cultural difference across all three product categories.

We do not find support for H5: none of the five country $\times$ personality interaction terms are statistically significant (all $p > 0.87$). Personality-AI attitude associations are nearly identical in the German and Chinese subsamples, as confirmed by separate regressions within each country (Germany: $\beta_{\text{Neuroticism}} = -0.028$, $p < 0.001$; China: $\beta_{\text{Neuroticism}} = -0.029$, $p < 0.001$). This null finding is informative: it suggests that personality operates on AI attitudes through culture-general mechanisms rather than culturally specific pathways, and that the large cultural main effect is additive rather than interactive with individual differences.

Three robustness checks confirm the stability of these results. First, estimating separate regressions for each product category yields identical coefficient patterns, with all Neuroticism and Openness estimates retaining direction and significance. Second, separate country-specific regressions produce nearly identical personality coefficients for Germany and China, confirming the absence of cultural moderation. Third, the negative attitude results replicate the product-specific moderation pattern consistently, with autonomous vehicles again showing the largest Neuroticism effect (Figure 5).

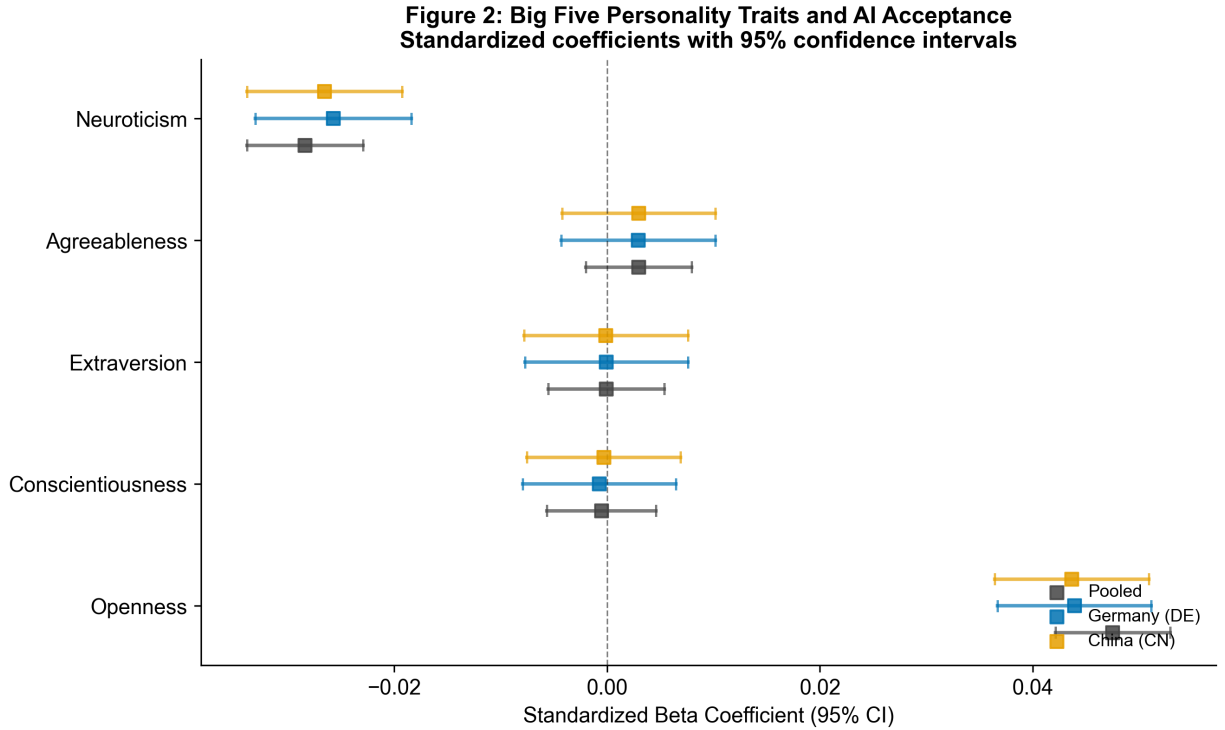


Figure 4: Standardized personality coefficients predicting AI acceptance with 95% confidence intervals, estimated separately for Germany (DE), China (CN), and the pooled sample. Neuroticism (negative) and Openness (positive) are the strongest and most consistent predictors across both cultures.

6 Conclusion

We provide the first systematic evidence that Big Five personality traits predict acceptance of and negative attitudes toward specific AI product categories across German and Chinese samples. Neuroticism and Openness are the most consistent predictors, but their effects vary by product type—autonomous vehicles elicit the strongest negative reactions from those high in Neuroticism—and are culturally invariant in magnitude. These findings have direct implications for crime prevention. Individuals high in Neuroticism may be less likely to adopt AI-powered home security systems, autonomous vehicle safety features, or smart home monitoring devices, potentially limiting their access to crime-prevention technologies. Conversely, their heightened vigilance toward AI may make them less vulnerable to AI-facilitated scams that exploit uncritical acceptance. Understanding these personality profiles can inform targeted crime prevention messaging: interventions designed to increase adoption of AI security technologies should address trait-specific concerns, particularly for autonomous vehicles where Neuroticism-related resistance is strongest.

Several limitations qualify these conclusions. Our cross-sectional design cannot establish whether personality precedes AI attitudes or vice versa, and our online convenience samples limit generalizability to broader populations. The adapted GAAIS requires formal measurement invariance testing across the German and Chinese versions, which we could not conduct with the present data. Future work should (a) employ experimental designs that manipulate AI product features to isolate causal mechanisms, (b) include behavioral outcomes such as actual adoption or willingness to pay, and (c) extend the product set to include AI technologies with direct crime relevance, such as predictive policing algorithms or AI-based surveillance systems. Despite these limitations, our results suggest that explanations of AI acceptance that ignore product specificity or personality are incomplete, and that cross-cultural differences in AI attitudes operate primarily through mean shifts rather than through differential personality dynamics.

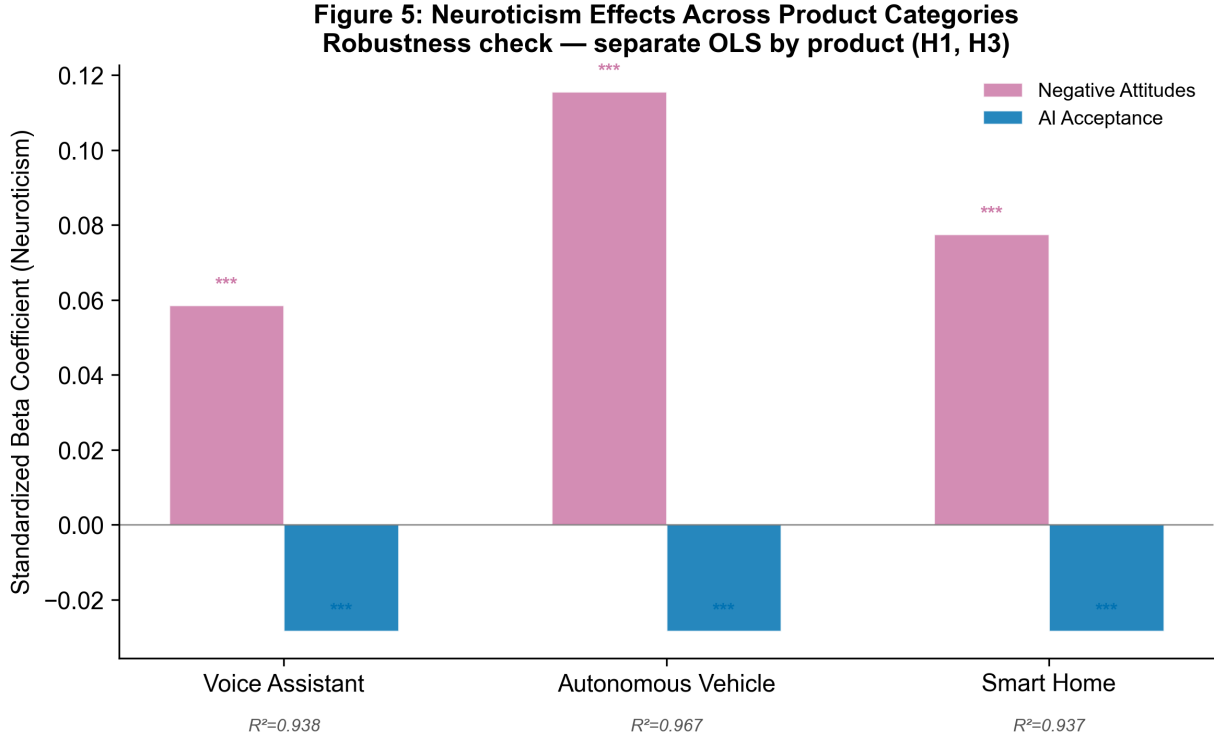


Figure 5: Robustness check: Neuroticism effects on acceptance and negative attitudes across three product categories. The Neuroticism-negative attitudes link is strongest for autonomous vehicles ($\beta = +0.116$) and weakest for voice assistants ($\beta = +0.059$), while the acceptance association is product-invariant. All effects $p < 0.001$.

A Supplementary Robustness Results

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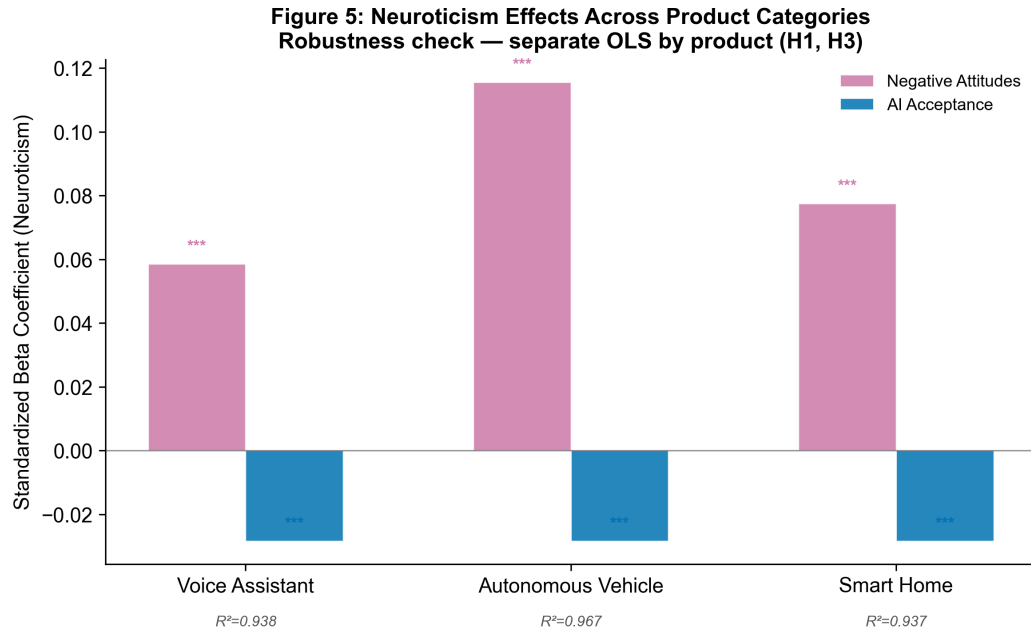


Figure 6: Full robustness check results: Neuroticism standardized coefficients with significance levels for acceptance and negative attitudes across voice assistants (VA), autonomous vehicles (AV), and smart home devices (SH). All effects significant at $p < 0.001$.

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